GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS

Project Report

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Technologies Used: Tableau, Python (Flask), HTML

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1. Project Overview

This project analyzes Olympic history using data visualization. It presents insights into medal distribution, country participation, and sports participation using Tableau charts, a dashboard, and a story. A Flask-based webpage integrates the published Tableau visualization.

2. Objectives

- Analyze Olympic medal distribution.
- Compare participation records across countries.
- Identify sports with the highest participation.
- Present insights through a Tableau dashboard and story.
- Integrate the visualization into a webpage using Flask.

3. Dataset and Data Preparation

The project uses the **120 Years of Olympic History — Athletes and Results** dataset from Kaggle.

The dataset contains information about athletes, countries, Olympic years, sports, events, and medals.

The data was used to create visualizations based on record counts. Top 10 filters were applied to country and sport charts to make comparisons easier to understand.

Dataset: https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results

4. Data Visualization

Three visualizations were created in Tableau:

1. **Medal Analysis:** Displays the distribution of Gold, Silver, and Bronze medal records.
2. **Olympic Participation by Country:** Shows the top 10 countries/NOCs by participation-record count.

3. **Sports Participation:** Shows the top 10 sports by participation-record count.

These visualizations help present Olympic data in a simple and understandable format.

5. Dashboard and Story

The Tableau dashboard combines the three visualizations into one view for easier comparison and analysis.

A Tableau Story titled **Olympic Performance Story** was also created to present the analysis in a structured sequence.

6. Web Integration

A webpage was developed using Python Flask and HTML. The application displays a project heading and an embedded Tableau visualization published on Tableau Public.

The main application file is app.py, and the webpage is stored in templates/index.html.

The application was tested locally using Flask.

7. Performance Testing

The following checks were performed:

- Verified that the Tableau worksheets displayed correctly.
- Checked dashboard layout and chart readability.
- Navigated through the Tableau Story.
- Tested the Flask webpage locally.
- Verified the embedded Tableau visualization on the webpage.

8. Conclusion

This project demonstrates how historical Olympic data can be explored using Tableau and presented through a Flask-based webpage. The visualizations provide an overview of medal distribution and participation patterns across countries and sports.

The project combines data analysis, visualization, and web integration to communicate information effectively.

9. References

1. Kaggle — 120 Years of Olympic History: https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results
2. Tableau Public: https://public.tableau.com/

3. Flask Documentation: https://flask.palletsprojects.com/